

Transmission Parameter Optimization of MIMO VLC System Assisted by IRS Based on DRL

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Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

With the rapid development of wireless communication technology, there is a growing demand for ultra-large capacity in communication systems. To satisfy this demand, the communication system not only widely adopts various key technologies to improve the spectral efficiency of radio frequency (RF) communication, such as large-scale multiple-inputs multiple-output (MIMO) and non-orthogonal multiple access (NOMA), but also builds a new physical layer architecture which include visible light, millimeter wave (mmWave) and terahertz (THz) frequency bands as an improvement and supplementary technology for RF communication [1].

Since Gfeller and Bapst [2] did the groundbreaking work of optical wireless communication (OWC), more and more interests were brought up because of its development prospects. In recent years, the pace of OWC progress has been very fast that visible light communication (VLC) which based on using prevalent light emitting diode (LED) lighting devices has been a valuable OWC technology and gained much attention. While existing wireless technologies are depleting the electromagnetic (EM) spectrum available in RF bands, VLC utilizes unregulated, undeveloped, and large EM bands in the visible range, which means that VLC has the advantage of being free of spectrum licensing. In addition to the above advantage, VLC has many favorable conditions, such as broad bandwidth, less interference with RF communications, excellent ability for reusing high frequency [4] and higher power efficiency [3]. Meanwhile, MIMO technology have been considered to achieve high data rates and reliability without consuming additional time and spectrum resources of VLC system [5] [6]. But visible light channels normally follow the typical Lambertian model, so that the channel gain depends on the position of the transceiver. In the MIMO VLC system, when the receiving photodiode (PD) distance is close, the MIMO channels have a serious correlation and the channel matrix condition number is too large to obtain the ideal multiplexing

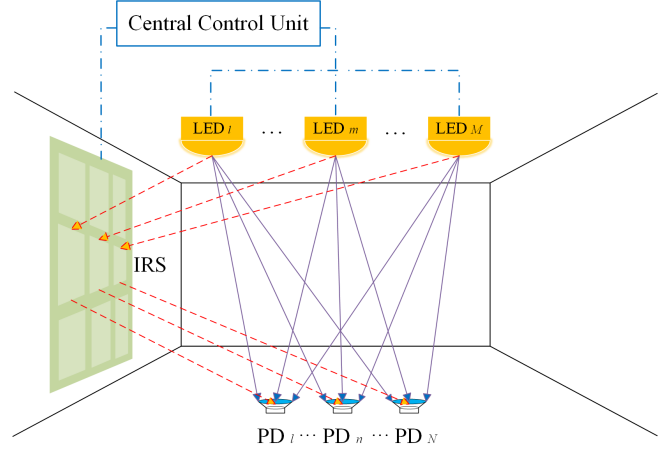


Fig. 1: Tunable IRS-Aided MIMO VLC System.

gain [7].

In order to solve above problems and increase the channel capacity of the MIMO VLC system, different kinds of PDs separately named pyramid receiver (PR) and hemispheric receiver (HR) were proposed to improve channel capacity and bit error rate (BER) performance [8]. In paper [9], the correlation of MIMO VLC channels is detailed in by singular value decomposition (SVD) of the signal of 4×4 MIMO VLC system, and the performance of the MIMO VLC system is improved by adaptive power and bits allocation strategy.

Since the communication of the VLC system mainly depends on the line of sight (LoS), and the signal gain of non-line of sight (NLoS) is negligible, the communication link will be interrupted and the communication situation will be affected once the LoS link is blocked. Nowadays, the emerging intelligent reflecting surface (IRS) start owning the capability of reconfiguring the wireless electromagnetic environment. Optical IRS overcomes the above shortcomings by improving the channel gain of VLC by providing an NLoS optical signal transmission path. Through controlling the relevant parameters of the IRS, the VLC system can surmount the high dependence of LoS, enhance the received signal power, expand the communication range, and relax the dimming constraints on the basis of meeting the lighting needs of indoor users [7]. In paper [10], it studied the most helpful mirror placement under

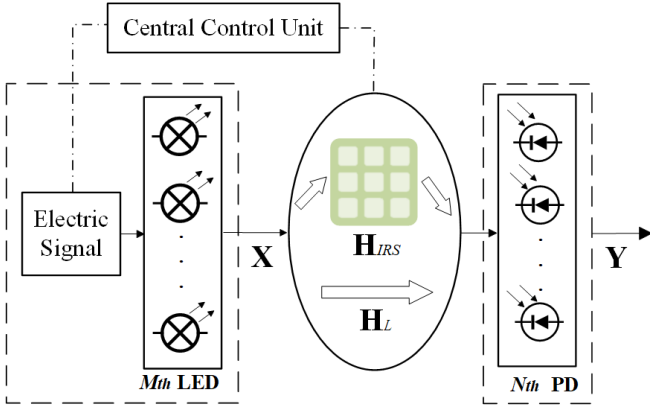


Fig. 2: IRS-Aided MIMO VLC System Model.

different conditions that can optimize the capacity of MIMO VLC system. Furthermore, in paper [11], it discussed that designing the IRS and processing transceiver signal are jointly considered to minimize the mean square error (MSE) of the IRS-aided MIMO VLC system with proposing an alternating optimization algorithm.

However, the above papers are all about IRS-aided MIMO VCL system with fixed reflection coefficients. Recently, with the development of optical metasurface technology [12], the sensitivity and response speed of the IRS to visible light have been improved quickly and steadily [13] [14]. An innovative IRS element with adjustable phase and amplitude is proposed, which also has the function of reflected signal power control [15]. In paper [16], the author studied resource management in IRS-aided MIMO VCL system based on deep-Q-learning algorithm as one type of deep reinforcement learning (DRL), which maximized the spectral efficiency (SE).

In this paper, we aims at optimizing the pairing matrix for the IRS and maximizing the capacity of IRS-aided MIMO VCL system. And in this system, the reflection coefficient is adjustable and waiting to be optimized. In paper [17], it modeled IRS-reflected channel and characterize the IRS-aided MIMO VLC capacity under different transmitter emission power constraints. We will use the capacity of the system as a maximization function as characterized above. The main contributions in our works are summarized as follows:

- First, we established a model of IRS-aided MIMO VCL system as shown in Fig.1, which the visible light channels follow the typical Lambertian model. Based on the above, we construct a general channel model between the signal transmitter and receiver, which is represented by $\mathbf{H}$, and general channel model also follow the corresponding dimming power constraints and pairing relationships between the LED, the IRS and the PD.
- Next, we put forward an optimization issues which aims at maximize system capacity characterized in [17]. Remarkably, because the parameters to be optimized include the reflection coefficient of the IRS, the LED power constraint, and the pairing relationship between

the IRS, the LED and the PD, which is a non-convex mixed integer optimization problem, it is difficult to solve this problem directly. Therefore, based on DRL, we establish a markov decision process (MDP), and use deep deterministic policy gradient (DDPG) algorithm to find the optimal solution when the system capacity is maximum.

- The final results show that the algorithm based on DDPG to maximize the system capacity are effective and convergent, meaning that the calculational complexity is substantially decreased and more efficient and equitable resource allocation scheme can be solved for IRS-aided VLC system.

The rest of this article consists of the following. In Section II, a spatial system and channel model is constructed, and in Section III we proposed a DDPG algorithm based on DRL to solve the optimal solution. Section IV provides numerical results to evaluate the performance of the proposed algorithm in the IRS-aided MIMO VLC system. In the end, Section V summarizes the article.

II. MODEL AND PROBLEM FORMULATION

A. System Model

Consider an IRS-assisted MIMO VLC system spatial model, as shown in Fig. 1. There are M LEDs, N PDs, and K IRS components in the system. By changing the reflection coefficient of the IRS element, the spatial freedom degree of the MIMO system can be improved, and the system capacity can be further improved. That is, the IRS-aided MIMO VLC system can change the channel matrix and reduce the correlation of the MIMO channel through changing the IRS reflection.

B. Channel Model

Consider an $M \times N$ dimensional IRS-aided MIMO VLC system, as shown in Fig. 2. The reflection coefficients of K IRS elements in the system are represented by the vector $\mathbf{r} = (r_1, \dots, r_K)$, $0 \leq r_k \leq \alpha$, and the r_k represents the maximum value of the reflection coefficient of the IRS elements. When the k -th IRS element is operating, the central control unit controls the intensity of the reflected light from the IRS element by setting the r_k .

The vector of the transmitted signal of M LEDs can be represented by $\mathbf{X} = (X_1, X_2, \dots, X_M)^T$. There are N PDs receiving signals at the same time. The received signal vector is $\mathbf{Y} = (Y_1, \dots, Y_N)^T$, and the $\mathbf{X}$ have the following relationship:

$$\mathbf{Y} = \mathbf{H}\mathbf{X} + \mathbf{Z} \quad (1)$$

where the $\mathbf{Z} = (Z_1, \dots, Z_N)^T$ is an additive gaussian white noise vector, namely $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$.

In the presence of IRS, the channel matrix includes the direct path gain $\mathbf{H}_L$ of the LED and the reflected path gain $\mathbf{H}_{IRS}$ of the IRS element. Thus, the channel matrix can be expressed as $\mathbf{H} = [\mathbf{H}_L + \mathbf{H}_{IRS}]$, $\mathbf{H} \in \mathbb{R}^{N \times M}$.

The gain of direct path can be expressed as

$$\mathbf{H}_L = [h_{nm}^{(L)}]_{N \times M} \quad (2)$$

$$h_{n,k,m}^{(IRS)} = \begin{cases} \frac{(\kappa+1)A_{PD}\cos^\kappa(\phi_{m,k}^{LR})\cos(\psi_{k,n}^{RD})\cos(\theta_{k,n}^{RD})}{2\pi(d_{kn}+d_{km})^2(\vec{N}_k^{IRS})^T\vec{E}_{k,n}^{RD}}T_s(\psi_{k,n}^{RD})g(\psi_{k,n}^{RD}), & 0 < \psi_{k,n}^{RD} < \Psi_C \\ 0, & \psi_{k,n}^{RD} > \Psi_C \end{cases} \quad (5)$$

where $h_{nm}^{(L)}$ represents the LoS channel gain from the m th LED light source to the n th PD, which can be calculated from the relative position between the PD and the LED:

$$h_{nm}^{(L)} = \begin{cases} \frac{(\kappa+1)A_{PD}}{2\pi d_{nm}^2}\cos^\kappa(\phi_{n,m})g(\psi_{n,m})T(\psi_{n,m})\cos(\psi_{n,m}), & 0 < \psi_{n,m} < \Psi_C \\ 0, & \psi_{n,m} > \Psi_C \end{cases} \quad (3)$$

where, A_{PD} is the detection area of the receiving PD, $\kappa = -1/\log_2(\cos \Phi_{1/2})$ is the Lambertian radiation order of the LED, the $\Phi_{1/2}$ is the half-radiation angle of the light source, d_{nm} represents the distance between the LED and the PD, $\phi_{n,m}$ is the radiation angle from the m th LED light source to the n th PD, $\psi_{n,m}$ is the angle of incidence from the m th LED light source to the n th PD, Ψ_C is the half-field of view of the PD, $g(\psi_{n,m})$ is the gain of the optical filter. The $T(\psi_{n,m})$ is the gain of the optical concentrator, which can be calculated by the following formula:

$$T(\psi_{n,m}) = \begin{cases} \frac{n_r^2}{\sin^2 \Psi_C}, & 0 < \psi_{n,m} < \Psi_C \\ 0, & \psi_{n,m} > \Psi_C \end{cases} \quad (4)$$

where n_r denotes the refractive index of the optical concentrator.

Regardless of the effect of the reflection coefficient, the channel gain reflected by the k th IRS element from the m th LED to the n th PD is expressed in $h_{n,k,m}$ and can be calculated by the formulation (5) at the top of this page. In formulation (5), d_{kn} and d_{km} respectively represent the distance from the k th IRS element to the n th PD and the m th LED light source, $\phi_{m,k}^{LR}$ represent the radiation angle from the m th LED light source to the k th IRS element, the $\theta_{k,n}^{RD}$ represents the radiation angle from the k th IRS element to the n th PD, $\psi_{k,n}^{RD}$ represents the angle of incidence from the k th IRS element to the n th PD, $T_s(\psi_{k,n}^{RD})$ and $g(\psi_{k,n}^{RD})$ still represent the optical concentrator gain and optical filter gain of the PD, $\vec{N}_k^{IRS}$ represents the unit normal vector of the k -th reflector surface and $(\vec{N}_k^{IRS})^T$ represents the rank of the vector, $\vec{E}_{k,n}^{RD}$ represents the directional unit vector from the k th IRS element to the n th PD. $\mathbf{H}_{IRS}$ can be represented in terms of binary matrices $\mathbf{G} = [\mathbf{g}_1, \dots, \mathbf{g}_M] \in \{0, 1\}^{K \times M}$ and $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N] \in \{0, 1\}^{K \times N}$, where $\mathbf{G}$ is the pairing between the IRS component and the LED, $\mathbf{F}$ is the pairing between the IRS element and the PD. Since the beam width of the reflected light of the IRS element is highly spatially limited, there is no crosstalk between the IRS reflection channels and each IRS element can only provide a reflection link for a pair of LED and PD. The binary matrix $\mathbf{G} \in \{0, 1\}^{K \times M}$

is used to represent the pairing between the IRS element and the LED,

$$\mathbf{G} = [\mathbf{g}_1, \dots, \mathbf{g}_M] = \begin{bmatrix} g_{1,1} & \cdots & g_{1,M} \\ \vdots & \ddots & \vdots \\ g_{K,1} & \cdots & g_{K,M} \end{bmatrix} = [\tilde{\mathbf{g}}_1, \dots, \tilde{\mathbf{g}}_K]^T \quad (6)$$

where $g_{k,m} = 1$ indicates that the k th IRS element reflects the VLC signal emitted by the m th LED, and $\mathbf{g}_m$ and $\tilde{\mathbf{g}}_k$ respectively represent the m th column and k th row of $\mathbf{G}$. The binary matrix $\mathbf{F} \in \{0, 1\}^{K \times N}$ is used to represent the pairing between the IRS element and the receiving PD, which can be expressed as

$$\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N] = \begin{bmatrix} f_{1,1} & \cdots & f_{1,N} \\ \vdots & \ddots & \vdots \\ f_{K,1} & \cdots & f_{K,N} \end{bmatrix} \quad (7)$$

where $f_{k,n} = 1$ means that the reflected light of the k th IRS element is received by the n th PD, and $\mathbf{f}_n$ means that the n th PD receives the VLC signal reflected by the IRS element. When $g_{k,m} = 1$ and $f_{k,n} = 1$, the k th IRS elements reflects the light signal from the m th LED to the n th PD. When considering the reflection coefficient of the IRS elements, the channel gain $\mathbf{H}_{IRS} \in \mathbb{R}^{N \times M}$ through the reflection path of the IRS element can be expressed as

$$\mathbf{H}_{IRS} = [(\mathbf{f}_n \odot \mathbf{g}_m)^T (\mathbf{r}^T \odot \mathbf{h}_{n,m})]_{N \times M} \quad (8)$$

where $\mathbf{h}_{n,m} = [h_{n,1,m}^{(IRS)}, \dots, h_{n,K,m}^{(IRS)}]^T$ represents the gain of the reflection path of K IRS elements between the m th LED and the n th PD without considering the reflection coefficient of the IRS element. $\odot$ denotes the Hadamard product, the multiplication of the corresponding elements in a matrix or vector. Therefore, we can define the channel matrix $\mathbf{H} = [\mathbf{H}_L + \mathbf{H}_{IRS}] = [h_{n,m}^{total}]_{N \times M}$, which the $h_{n,m}^{total}$ can be expressed as

$$h_{n,m}^{total} = h_{n,m}^{(L)} + (\mathbf{f}_n \odot \mathbf{g}_m)^T (\mathbf{r}^T \odot \mathbf{h}_{n,m}) \quad (9)$$

Since each IRS element can only provide reflections for a pair of LEDs and PDs, $g_{k,m}$ and $f_{k,n}$ need to meet the constraints:

$$\sum_{m=1}^M g_{k,m} \leq 1, \forall k \in \mathcal{K} \quad (10)$$

$$\sum_{n=1}^N f_{k,n} \leq 1, \forall k \in \mathcal{K} \quad (11)$$

In addition, the emitted light intensity of the LEDs light source needs to meet the peak power constraint and the dimming constraint:

$$0 \leq X_m \leq A_m, \quad m = 1, \dots, M \quad (12)$$

$$E[X_m] = E_m, \quad m = 1, \dots, M \quad (13)$$

where $\mathbf{A} = [A_1, \dots, A_M]^T$, $A_m > 0$ represents the maximum luminous intensity of M LEDs, and $\mathbf{E} = [E_1, \dots, E_M]^T$, $E_m > 0$ represents the average light intensity of the LEDs.

C. Problem Formulation

In [12], it is stated that the progressive capacity of an IRS-aided MIMO VLC system can be given by the sum of the following common terms and a specific constant term:

$$f(\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r}) = \frac{1}{2} \log \det [\mathbf{H}^T (\sigma^2 \mathbf{I}_N)^{-1} \mathbf{H}] + \sum_{m=1}^M \log A_m \quad (14)$$

where $\sigma^2 \mathbf{I}_N$ is the variance matrix of the additive gaussian white noise.

Therefore, in order to maximize the capacity of the IRS-aided MIMO VLC system, it is equivalent to maximizing $f(\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r})$. Based on the above situation, the capacity maximization problem of the IRS-aided MIMO VLC system studied in this paper can be expressed as follows:

$$\begin{aligned} P: \max_{\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r}} & f(\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r}) \\ \text{s.t.} & f_{k,m}, g_{k,m} \in \{0, 1\}, \forall k \in \mathcal{K}, m \in \mathcal{M}, \\ & E_m > 0, \\ & 0 \leq r_k \leq \alpha, \\ & \sum_{m=1}^M g_{k,m} \leq 1, \forall k \in \mathcal{K}, \\ & \sum_{n=1}^N f_{k,n} \leq 1, \forall k \in \mathcal{K}, \\ & 0 \leq X_m \leq A_m, \quad m = 1, \dots, M, \\ & E[X_m] = E_m, \quad m = 1, \dots, M, \end{aligned} \quad (15)$$

It is important to note that Eq.(15) is difficult to obtain an optimal solution for the following reasons. First, the objective function $f(\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r})$ is non-concave. Secondly, due to the binary constraints in $f_{k,m}$ and $g_{k,m}$, integer programming is also involved. Therefore, we proposed that the transmission parameter configuration scheme to maximize the capacity of the system is solved by DRL. The MDP state is the position and channel estimation of the receiver, the action includes the possible value of the transmission parameters, the policy is the transmission parameter configuration scheme that satisfies the constraints, and the reward function is the capacity of the IRS-assisted MIMO VLC system.

III. DRL-BASED ALGORITHM

In order to formulate the optimal strategy using the DRL-based algorithm, we construct an MDP so that the DDPG algorithm can be applied to solve Eq.(15). It is assumed that the IRS-aided MIMO VLC system is the environment and the central control unit is the agent.

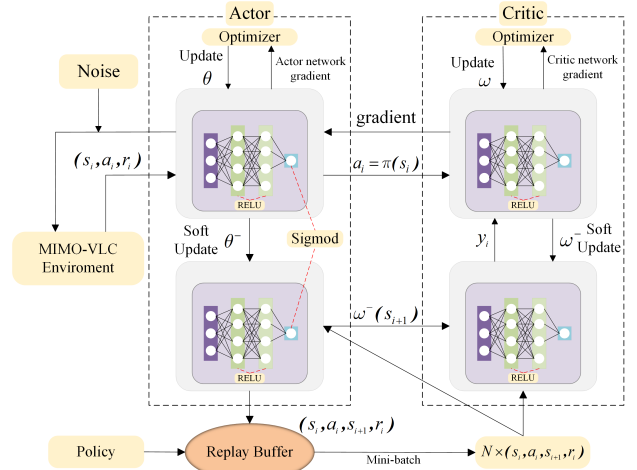


Fig. 3: DDPG principle

A. MDP Model

In this MDP model, the state space, action space, reward, and state-action value function can be represented as follows:

- State space: We formulate the state space vector s_t as

$$s_t = [\text{vec}(\mathbf{G}), \text{vec}(\mathbf{F}), \text{vec}(\mathbf{A}), \text{vec}(\mathbf{r}), \text{vec}(\mathbf{H})] \quad (16)$$

where $\text{vec}(\cdot)$ is a vectorization operator. The dimension of the state space is $D_{state} = (M+N)K + M + K + NM$.

- Action space: Given that our goal is to obtain the optimal IRS correlation matrix, reflection coefficient vector and power allocation vector, we formulate the action vector a_t as:

$$a_t = [\text{vec}(\Delta \mathbf{G}), \text{vec}(\Delta \mathbf{F}), \text{vec}(\Delta \mathbf{A}), \text{vec}(\Delta \mathbf{r})] \quad (17)$$

where the dimension of the action space is $D_{action} = (M+N)K + M + K$. Noteworthily, the matrices $\mathbf{F}$ and $\mathbf{G}$ in the above formula are all binary, that means they need to be relaxed.

- Reward: Maximizing the capacity of the IRS-aided MIMO VLC system is the goal of the optimization problem, so the reward function r_t is defined as the capacity of the system in the current state:

$$r_t = f(\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r}) = \frac{1}{2} \log \det [\mathbf{H}^T (\sigma^2 \mathbf{I}_N)^{-1} \mathbf{H}] + \sum_{m=1}^M \log A_m \quad (18)$$

- State-action value function: Define $Q^\pi(\cdot)$ as state-action value function when the policy executes an action in the state. It can be expressed as follow on the basis of the Bellman equation,

$$Q^\pi(s_t, a_t) = E_\pi[r_t + \varepsilon Q^\pi(s_{t+1}, a_{t+1}) | s_t, a_t] \quad (19)$$

where $\varepsilon < 1$ is defined as the reward discount factor for the next state.

B. The DDPG-Based Algorithm

The basic block diagram of DDPG is shown in Fig.3 and the process is shown in Algorithm 1. In the structure of the DDPG algorithm, there are four main networks which named actor network, critic network, target actor network and target critic network. The actor network and the critic network are updated respectively by the backpropagation algorithm (BP) by parameters θ and ω , while the target actor network and the target critic network are updated by θ^- and ω^- . We extract the current state from the environment and input it into the actor network to get the action, and input it into the critic network to generate a value function to evaluate the current action state function.

In the actor network, the dimension of the input layer is D_{state} , and the dimension of the output layer is D_{action} , and the Sigmoid function is used as the activation function between layers. There are two hidden layers, and the number of neurons in each hidden layer is 880, and the activation function is set to the Relu function. Each layer is fully connected to each other, and the output of each layer is normalized to improve learning performance and reduce the impact of initial values[18].

In the critic network, the dimension of the input layer is $D_{action} + D_{state}$ and only one dimension in the output layer. This is because the critic network needs to input states and actions, and output a state-action value to evaluate the current policy. There are two hidden layers, and the number of neurons in each hidden layer is 880. The activation functions, output normalization, and connections between layers are consistent with the actor network. The Adma optimizer is used for both actor network and also for critic network to update.

During the training and updating of neural networks, what we do first is to store the sample to replay buffer. After this, we draw N samples from the replay buffer and calculate the target value function:

$$y_i = r_i + \gamma Q_{\omega^-}(s_{i+1}, \mu_{\theta^-}(s'_i)) \quad (20)$$

Next, we minimize the loss function as below and update the critic network with it.

$$L = \frac{1}{N} \sum_{i=1}^N (y_i - Q_{\omega}(s_i, a_i))^2 \quad (21)$$

And the update of the action network parameters is determined by

$$\nabla_{\theta} J \approx \frac{1}{N} \sum_{i=1}^N \nabla_{\theta} \mu_{\theta}(s_i) \nabla_a Q_{\omega}(s_i, a) \big|_{a=\mu_{\theta}(s_i)} \quad (22)$$

Finally, the target actor network and the target critic network are softly updated. Particularly, the $\mathbf{F}$ and $\mathbf{G}$ matrices need to satisfy binary, but since the effect of noise on the action cannot make them absolutely binary, we binarize the $\mathbf{F}$ and $\mathbf{G}$ matrices after adding noise so that their values are absolutely 1 and 0.

Algorithm 1: Joint Optimization based on DDPG

```

1: Initialize;
2: Actor network  $\mu_{\theta}(s)$  given  $\theta$ ;
3: Critic network  $Q_{\omega}(s, a)$  given  $\omega$ ;
4: Copy the same parameters  $\theta \rightarrow \theta^-$  and  $\omega \rightarrow \omega^-$  to
   initialize the target networks  $\mu_{\theta^-}$  and  $Q_{\omega^-}$  respectively.
5: Set replaybuffer  $\mathcal{R}$  with size  $|\mathcal{B}|$ ;
6: for  $episode = 1 \rightarrow E$  do do
7:   Initialize the Gaussian white noise  $\mathcal{N}$ ;
8:   Reset the initial state of the environment  $s_1$ ;
9:   for  $step = 1 \rightarrow T$  do do
10:    Select action  $a_t = \mu_{\theta}(s_t) + \mathcal{N}$  based on the current
       strategy and noise;
11:    Obtain  $(r_t, s_{t+1})$  from environment;
12:    Store the transition  $s_t, a_t, r_t, s_{t+1}$  into  $\mathcal{R}$ ;
13:    while  $|\mathcal{R}| > N$  do
14:      Sample  $N$  tuples  $e_i = \{(s_i, a_i, r_i, s_{i+1})\}_{i=1, \dots, D}$ 
       from  $\mathcal{R}$ ;
15:      For each tuple, using the target network to
       compute by (20);
16:      Minimize target losses by (21);
17:      Update the Critic network;
18:      Calculate the sampling policy gradient (22);
19:      Update the Actor network;
20:      Soft update the target networks by:
21:         $\omega^- \leftarrow \tau \omega + (1 - \tau) \omega^-$ ;
22:         $\theta^- \leftarrow \tau \theta + (1 - \tau) \theta^-$ ;
23:    end while
24:  end for
25: end for
26: return  $\mathbf{G}^*, \mathbf{F}^*, A^*, \mathbf{r}^*$ 

```

IV. SIMULATION AND RESULTS

A. Simulation scenarios

We set up a room of size $6m \times 6m \times 9m$ as shown in Fig.1 to simulate visible light communication. At the top of the room, we placed four LEDs in the middle as signal transmitters arranged evenly. Sixteen IRSs are set in the center on the side walls, which are arranged in a grid. Five PDs are placed at the bottom of the room, with a receiving area of $1cm^2$, which are arranged randomly. The rest of the simulation parameters are shown in the table below.

B. Simulation Results

REFERENCES

- [1] R. W. Heath, N. González-Prelcic, S. Rangan, W. Roh and A. M. Sayeed, "An Overview of Signal Processing Techniques for Millimeter Wave MIMO Systems," in IEEE Journal of Selected Topics in Signal Processing, vol. 10, no. 3, pp. 436-453, April 2016, doi: 10.1109/JSTSP.2016.2523924.
- [2] F. R. Gfeller and U. Bapst, "Wireless in-house data communication via diffuse infrared radiation," in Proceedings of the IEEE, vol. 67, no. 11, pp. 1474-1486, Nov. 1979, doi: 10.1109/PROC.1979.11508.
- [3] T. Komine and M. Nakagawa, "Fundamental analysis for visible-light communication system using LED lights," in IEEE Transactions on Consumer Electronics, vol. 50, no. 1, pp. 100-107, Feb. 2004, doi: 10.1109/TCE.2004.1277847.

- [4] M. Obeed, A. M. Salhab, M. -S. Alouini and S. A. Zummo, "On Optimizing VLC Networks for Downlink Multi-User Transmission: A Survey," in *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2947-2976, thirdquarter 2019, doi: 10.1109/COMST.2019.2906225.
- [5] K. -H. Park, Y. -C. Ko and M. -S. Alouini, "On the power and offset allocation for rate adaptation of spatial multiplexing in optical wireless MIMO channels," 2011 7th International Wireless Communications and Mobile Computing Conference, Istanbul, Turkey, 2011, pp. 141-146, doi: 10.1109/IWCMC.2011.5982521.
- [6] C. Chen, W. -D. Zhong, D. Wu and Z. Ghassemlooy, "Wide-FOV and High-Gain Imaging Angle Diversity Receiver for Indoor SDM-VLC Systems," in *IEEE Photonics Technology Letters*, vol. 28, no. 19, pp. 2078-2081, 1 Oct. 1, 2016, doi: 10.1109/LPT.2016.2584185.
- [7] S. Sun, T. Wang, F. Yang, J. Song and Z. Han, "Intelligent Reflecting Surface-Aided Visible Light Communications: Potentials and Challenges," in *IEEE Vehicular Technology Magazine*, vol. 17, no. 1, pp. 47-56, March 2022, doi: 10.1109/MVT.2021.3127869.
- [8] A. Nuwanpriya, S. -W. Ho and C. S. Chen, "Indoor MIMO Visible Light Communications: Novel Angle Diversity Receivers for Mobile Users," in *IEEE Journal on Selected Areas in Communications*, vol. 33, no. 9, pp. 1780-1792, Sept. 2015, doi: 10.1109/JSAC.2015.2432514.
- [9] Y. Zhai, H. Chi, J. Tong and J. Xi, "Capacity Maximized Linear Precoder Design for Spatial-Multiplexing MIMO VLC Systems," in *IEEE Access*, vol. 8, pp. 63901-63909, 2020, doi: 10.1109/ACCESS.2020.2983517.
- [10] B. G. Guzman, M. M. Cespedes, V. P. G. Jimenez, A. G. Armada and M. Brandt-Pearce, "Optimal Mirror Placement to Minimize the Outage Area in Visible Light Communication," *GLOBECOM 2023 - 2023 IEEE Global Communications Conference*, Kuala Lumpur, Malaysia, 2023, pp. 4698-4703, doi: 10.1109/GLOBECOM54140.2023.10436718.
- [11] S. Sun, F. Yang, J. Song and R. Zhang, "Intelligent Reflecting Surface for MIMO VLC: Joint Design of Surface Configuration and Transceiver Signal Processing," in *IEEE Transactions on Wireless Communications*, vol. 22, no. 9, pp. 5785-5799, Sept. 2023, doi: 10.1109/TWC.2023.3236811.
- [12] N. Ullah, R. Zhao, and L. Huang, "Recent Advancement in Optical Metasurface: Fundament to Application," *Micromachines*, vol. 13, no. 7, p. 1025, Jun. 2022.
- [13] T. He, T. Liu, S. Xiao, Z. WEI, Z. WANG, L. Zhou AND X Cheng, "Perfect Anomalous Reflectors at Optical Frequencies," in *Science Advances*, vol. 8, no. 9, 2022.
- [14] S. Abdollahramezani, O. Hemmatyar, M. Taghinejad, et al. "Electrically driven reprogrammable phase-change metasurface reaching 80% efficiency," in *Nature Communications*, vol. 13, 1696, 2022.
- [15] B. Chen, X. Wang, W. Li, Chun Li and Z. Wang, et al, "Electrically Addressable Integrated Intelligent Terahertz Metasurface, " in *Science Advances* vol. 8, no. 41, 2022.
- [16] A. A. Hammadi, L. Bariah, S. Muhaidat, M. Al-Qutayri, P. C. Sofotasios and M. Debbah, "Deep Q-Learning-Based Resource Management in IRS-Assisted VLC Systems," in *IEEE Transactions on Machine Learning in Communications and Networking*, vol. 2, pp. 34-48, 2024, doi: 10.1109/TMLCN.2023.3328501.
- [17] S. Sun, W. Mei, F. Yang, N. An, J. Song and R. Zhang, "Optical Intelligent Reflecting Surface Assisted MIMO VLC: Channel Modeling and Capacity Characterization," in *IEEE Transactions on Wireless Communications*, vol. 23, no. 3, pp. 2125-2139, March 2024, doi: 10.1109/TWC.2023.3295793.
- [18] D. Pereira-Ruisánchez, Ó. Fresnedo, D. Pérez-Adán and L. Castedo, "Deep Contextual Bandit and Reinforcement Learning for IRS-Assisted MU-MIMO Systems," in *IEEE Transactions on Vehicular Technology*, vol. 72, no. 7, pp. 9099-9114, July 2023, doi: 10.1109/TVT.2023.3249353.